Thakur Trusts (Regd.)
THAKUR COLLEGE OF
ENGINEERING & TECHNOLOGY
Autonomous College Affiliated to University of Mumbai
Approved by All India Council for Technical Education (AICTE) and Government of Maharashita(GOM)
Conferred Autonomous Status by University Grants Commission (UGC) for 10 years w.e.f. A.Y. 2019-20
(among the First in the country to be conferred Autonomous Status) and awarded 'A+' Grade by NAAC (3rd Cycle)
(Institute Ranked 58th in India by NIRF, Ministry of Education, Government of India)
Estd. 2001

IN-SEMESTER EXAMINATION-I

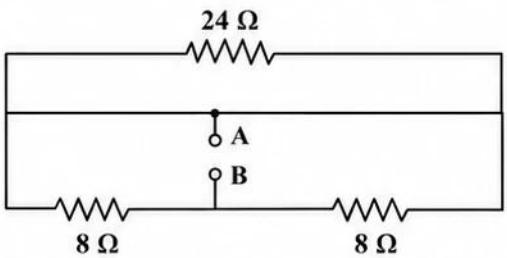
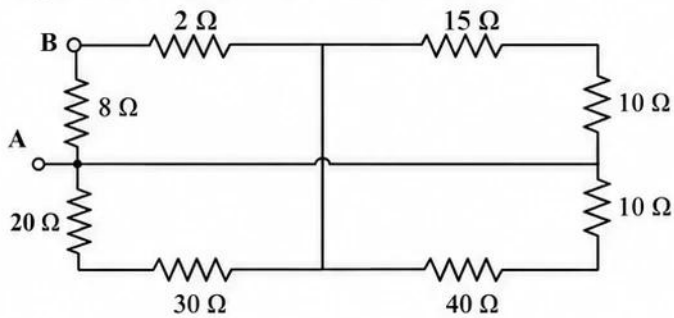
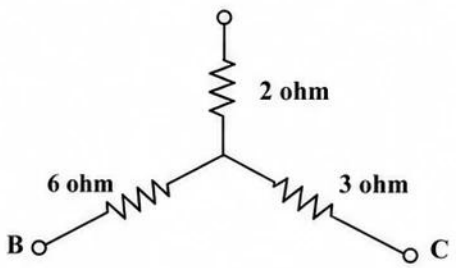
F.E./F.T. (Semester-I)

SUBJECT – Basic Electrical Engineering

Branch: COMP, CIVIL, E&CS, CSE, AI&DS, IoT **Date: 25/10/2024**
Div.: ALL **Timing: 10:00 AM to 11:00 AM**
Duration: 60 Minutes **Maximum Marks: 20**

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	Find the equivalent resistance between terminals A and B. 	2	1	A
	b.	Find the equivalent resistance between terminal A and B. 	2	1	A
	c.	If this star connected resistor is converted into Delta what will be the value of RAB. 	2	1	A

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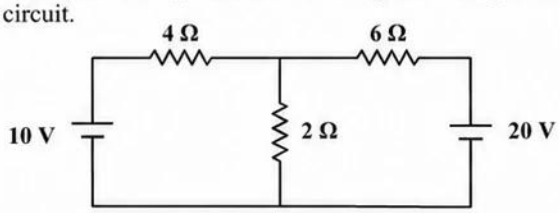
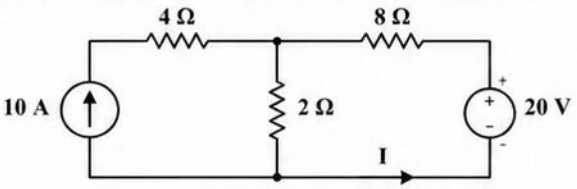
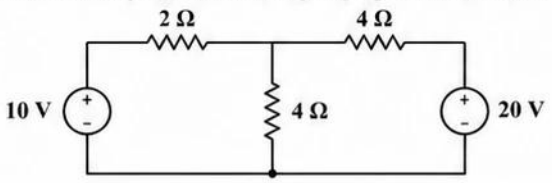
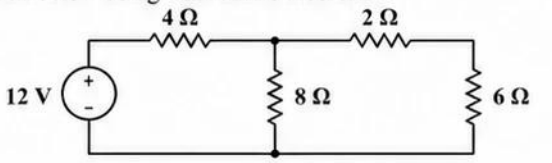
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Q. No.		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
Q.2	a.	Determine the current through 2 Ω resistor using Mesh analysis for the following circuit. 	5	2	A
		OR			
	b.	Find the current I using Nodal analysis for the following circuit. 	5	2	A
Q.3	a.	State and explain Superposition theorem. Find the current through 4 Ω resistor in the circuit given below using Superposition theorem. 	5	3	R
		OR			
	b.	State and explain Thevenin's theorem. Find the current through 6 Ω resistor in the circuit given below using Thevenin's theorem. 	5	3	R